Computational Machine Learning — Assignment 3

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Classifying Colorectal Cancer Cells from Histopathology Images

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0. Introduction

The aim is to address two classification problems on a modified version of the CRCHistoPhenotypes dataset, a collection of small histopathology image patches each centred on a single colorectal cell. The first task is binary that decides a cell is cancerous. The second is on four classes since each cell might be assigned to one of epithelial, inflammatory, fibroblast or other.

The work is organised around two ideas that turned out to dominate every result. The first is that the images carry genuine non-linear structure, so model capacity helps up to a point but no further. The important second is that the data is grouped by patient, and honest evaluation has to respect that grouping. Throughout the experiment I keep the difference that various models have on a validation set and how they generalize to patients they have never seen, because that gap determines the capacity of the model.

A. Task Definition

The two tasks share the same images but use different labels and different subsets of the data.

Task A — `isCancerous`. A binary label is available for every image, so this task uses the full dataset of 20,280 cells (the 9,896 fully-annotated "main" cells plus the 10,384 "extra" cells that carry only the cancer label). The classes are imbalanced, with roughly 13,200 non-cancerous against 7,070 cancerous cells.

Task B — `cellType`. The four-way cell-type label exists only for the main data, so this task uses the 9,896 main cells. The classes are more skewed: epithelial (4,079), inflammatory (2,543), fibroblast (1,888) and other (1,386).

In the main data the count of cancerous cells is exactly 4,079, which is also the count of epithelial cells, and inspection confirms these are the same cells. The cancerous cells are the epithelial cells. Task A and the epithelial class of Task B are describing the same biological object from two angles, which is why the epithelial class is the easiest of the four to be recognised.

The success criteria set for the assignment are a macro-averaged F1 of at least 0.90 for Task A and at least 0.60 for Task B, and measured on patients the model has never seen. Macro-F1 is the right measure here precisely because of the imbalanced classes, so it weights every class equally. Therefore, a model cannot score well simply by getting the majority classes right and ignoring the minor ones.

B. Dataset Description and Exploratory Data Analysis

Each image is a 27×27 RGB patch, which is small enough that a single cell fills most of the frame but leaves little room for surrounding context. The dataset draws on 98 patients in total, 60 in the main data and 38 in the extra data.

Class balance

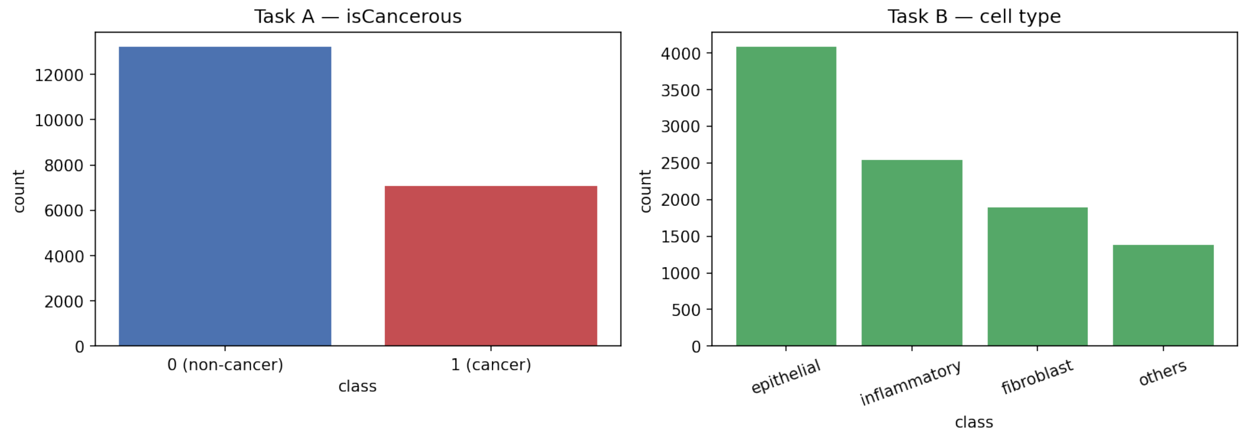

Figure 1: Class distribution for both tasks

Both tasks are imbalanced, and Task B: the rarest class, *other*, has barely a third of the examples of the most common, *epithelial*.

What the average cell looks like

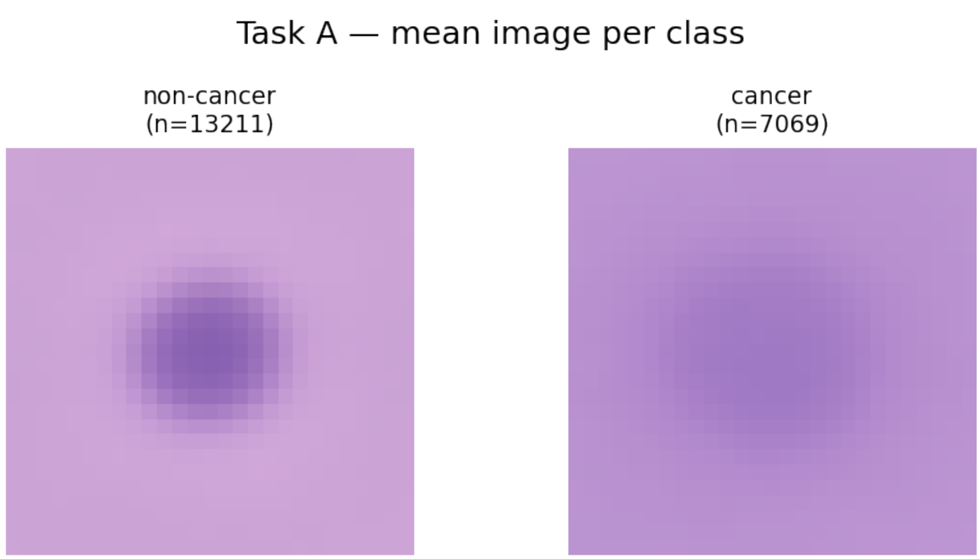

Figure 2: Mean image per class, Task A

Averaging every image in a class gives a rough sense of how separable the classes are by appearance alone. The two Task A means differ in a soft, diffuse way rather than in any sharp feature, which

Figure 3: Mean image per class, Task B

already warns that a simple model working on raw pixels will struggle. The Task B means are more distinct between epithelial and the rest, but fibroblast and other look very similar to each other.

Unsupervised structure

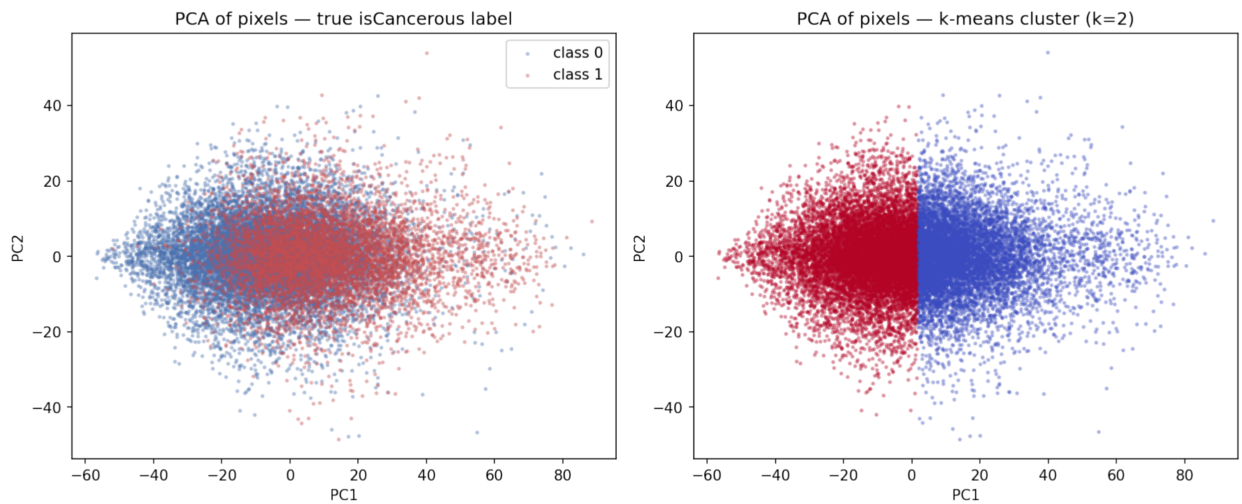

Figure 4: PCA projection coloured by k-means clusters

Appying the unsupervised methopd to check whether the classes form natural groups in pixel space I flattened each image to a 2,187-dimensional vector, reduced it with PCA, and ran k-means. The first two principal components capture only 0.184 and 0.042 of the variance, so most of the signal is spread thinly across many dimensions rather than concentrated in a few. Moreover, the k-means clustering agrees with the true cancer labels at an adjusted Rand index of only 0.115, which is close to chance. The classes do not sit in tidy.

EDA indicates that this is a prediction, not just a description. If the classes are not linearly separable in pixel space, then linear models and any classifier working on raw flattened pixels should do poorly, while a model that can learn its own non-linear, spatially aware features should do markedly better.

C. Data Pre-processing

Patient-level splitting

The most important pre-processing decision is how to split the data. Cells from the same patient are not independent since they come from the same tissue, were stained in the same batch and imaged on the same scanner, so they share visual parts that have nothing to do with the label. If cells from one patient appear in both training and test sets, the model can learn those patterns and the test score becomes a flattering measure of memory rather than an real evaluation.

To prevent that, I split by patient, not by image, using a grouped shuffle split into 60% training, 20% validation and 20% test, with the constraint that no patient appears in more than one set. The patient sets are disjoint for both tasks.

Feature scaling

Two representations are needed. The classical models and the MLP take flattened pixel vectors, which I standardise with a scaler fitted on the training data only. The convolutional networks take the images as $3\times27\times27$ tensors, normalised per colour channel using means and standard deviations again computed only on the training set. It only fits on training data.

D. Model Development and Performance Analysis

For each task I built three models to experiment the capacity, then tuned the strongest of them. The aim was not only to find the best classifier but to watch how performance goes, since the EDA gave a clear prediction about what that relationship should look like.

Classical baselines

A decision tree and logistic regression, both on standardised flattened pixels. The decision tree reached a validation macro-F1 of 0.729 on Task A and 0.427 on Task B, logistic regression did better at 0.799 and 0.492. Both are well short of the targets, when logistic regression can only draw linear boundaries, and the decision tree splits one pixel at a time, so neither can express the non-linear pixel interactions that the EDA showed are necessary.

Multilayer perceptron

A two-hidden-layer MLP with dropout lifts validation macro-F1 to 0.826 on Task A and 0.534 on Task B. The hidden layers let the model combine pixels in ways a logic regression cannot. But the MLP still treats the image as an unstructured list of pixels and throws away the fact that nearby pixels belong together.

Convolutional neural network and hyperparameter tuning

The CNN uses three convolution-and-pooling blocks followed by a small fully connected head. Its inductive bias matches the data: convolution looks at local neighbourhoods and shares weights across the image, so it can learn edge, texture and shape detectors that are exactly the features a pathologist uses.

I tuned the learning rate over {0.01, 0.001, 0.0001}, training each setting with an identical initialisation for a fair comparison.

Logistic Regression

Decision Tree

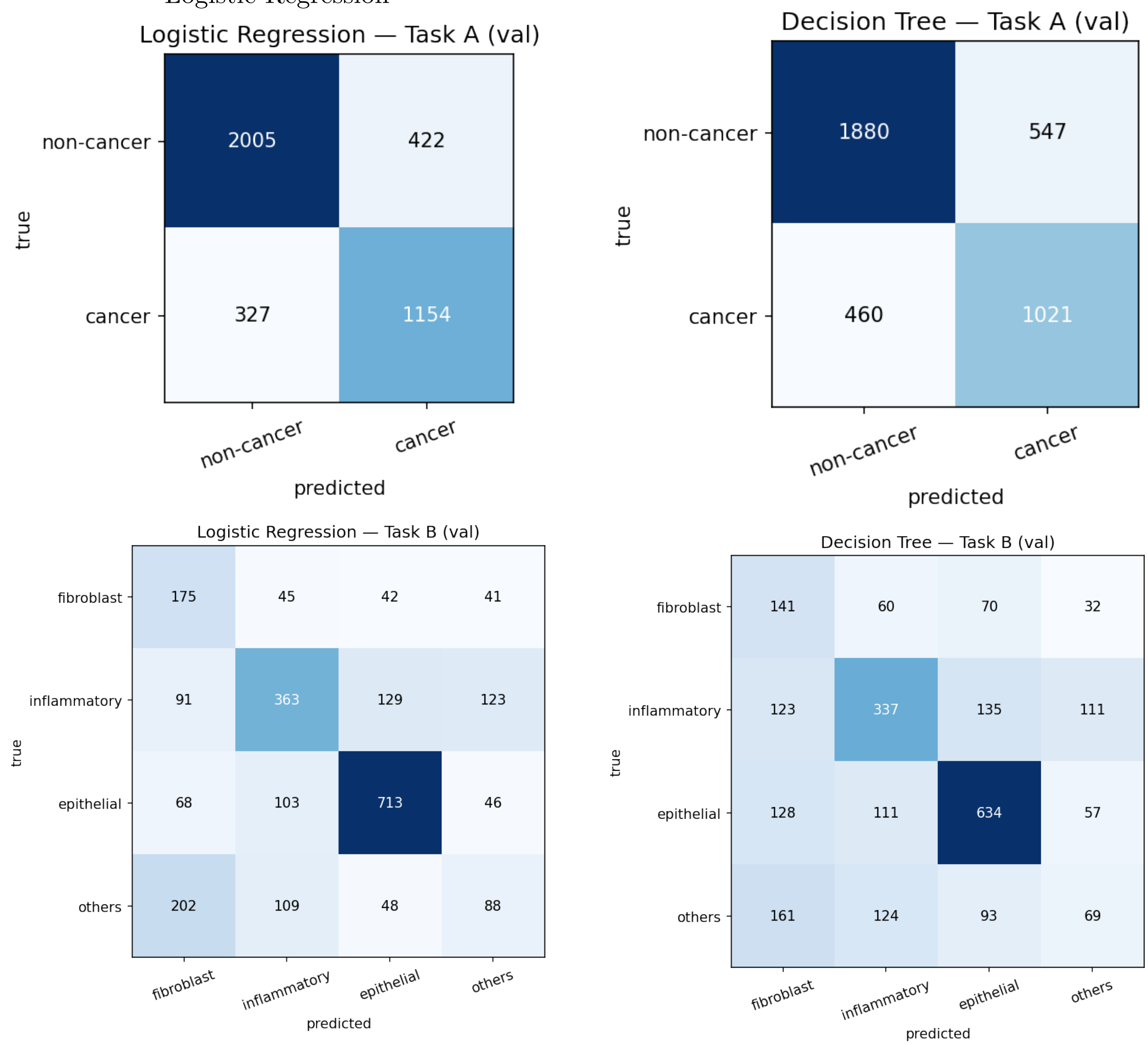

Figure 5: Classical-baseline confusion matrices (Task A top row, Task B bottom row). Both models lean on the common classes and confuse the rare ones (fibroblast against *other* in Task B), the expected result for linear and axis-aligned splits on raw pixels.

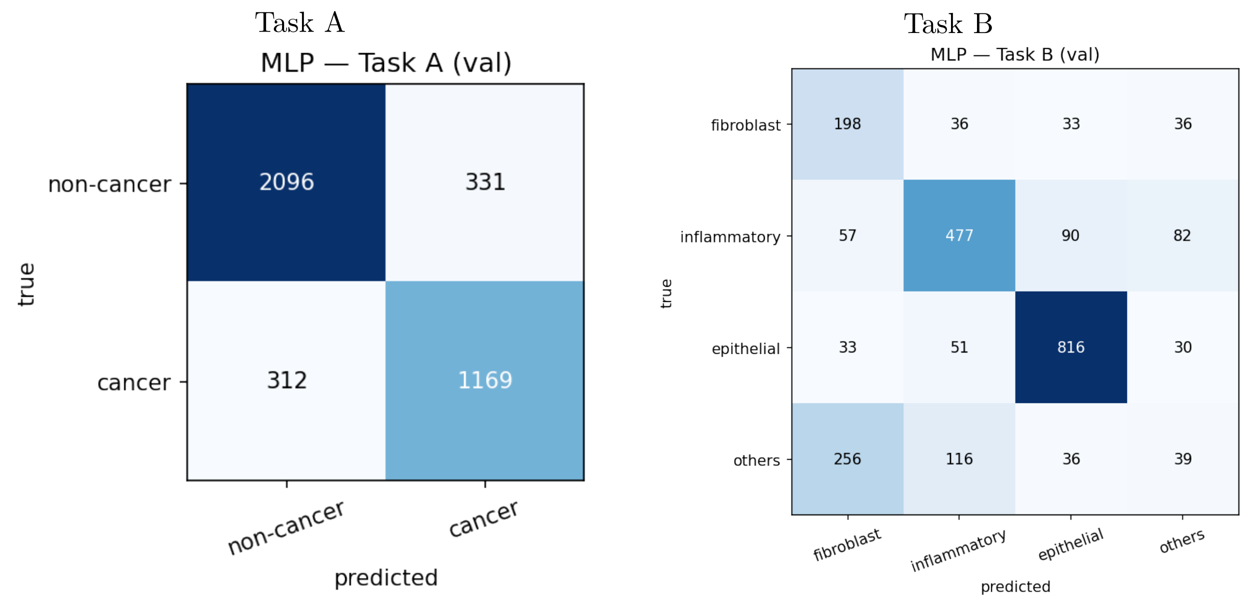

Figure 6: MLP confusion matrices. Non-linearity sharpens the common classes over the linear baselines, but the rare Task B classes (fibroblast, *other*) remain weak because the flattened input discards spatial structure.

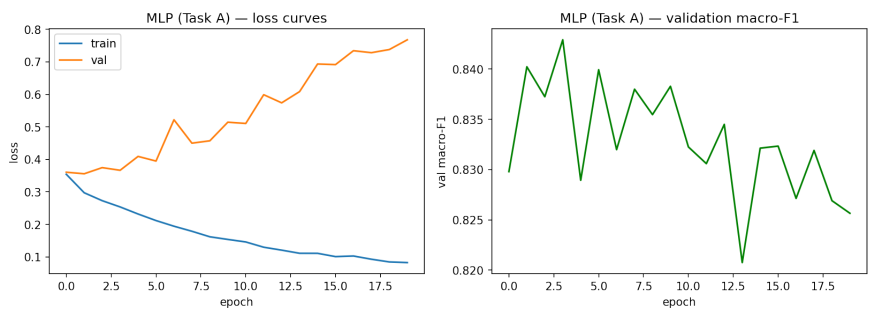

Figure 7: Task A MLP loss curves. Validation loss turns upward early while training loss keeps falling: the flat model overfits, foreshadowing the same diagnosis for the CNN and the need for the regularisation in Section E.

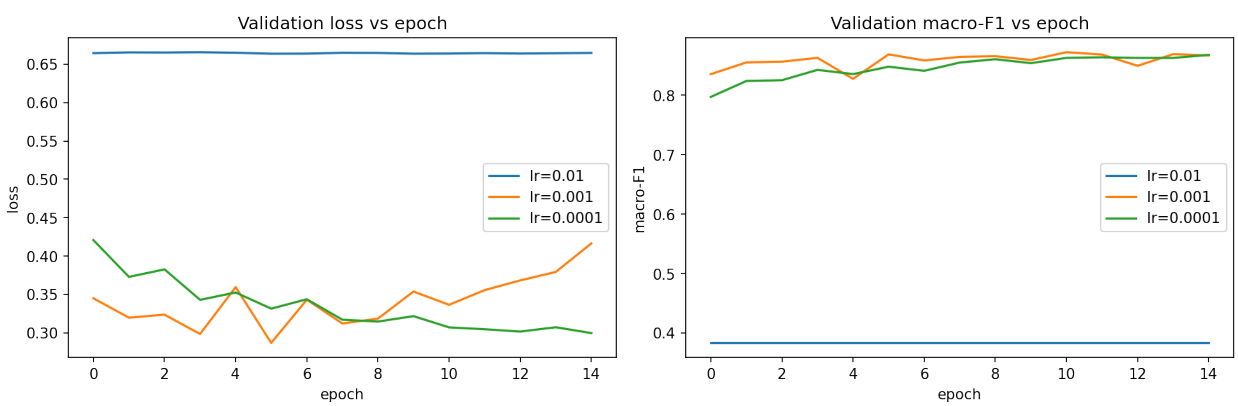

Figure 8: CNN learning-rate tuning on Task A

A rate of 0.01 was unstable and 0.0001 learned too slowly within centain epochs, 0.001 gave the best and steadiest validation macro-F1, so I used it for both tasks. The tuned CNN reached 0.867 on Task A and 0.635 on Task B, comfortably the best of the three model families and the first model to reach the Task B target.

The loss curves, however, revealed a problem.

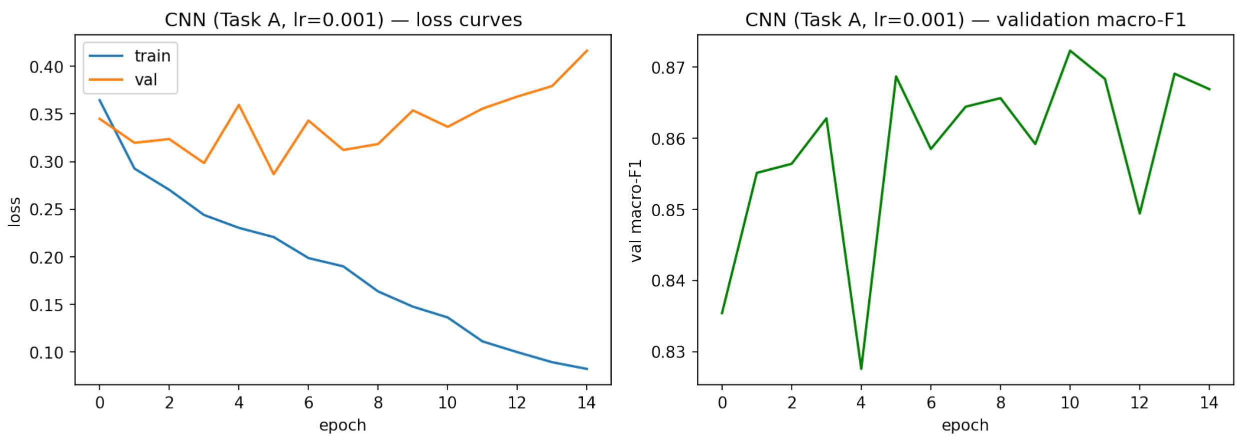

Figure 9: Task A CNN loss curves, showing overfitting

The training loss keeps falling while the validation loss bottoms out around epoch five and then drifts upward. The model is overfitting clearly when the network has enough capacity to start memorising the training patients rather than learning features that transfer.

E. Advanced Techniques

The assignment asks for at least two advanced techniques. I use data augmentation and a soft-voting ensemble. I initially planned transfer learning from CIFAR-10, but pre-training a backbone on natural photographs is computational. An ensemble reuses models I had already trained and is far cheaper, so it was the better fit.

E.1 Data augmentation

The plain CNN overfit, so the improvement is to enlarge the effective training set. Before each pass, every training image is randomly flipped horizontally and vertically, rotated by up to twenty degrees, and given a small brightness and contrast jitter. The network therefore almost never sees the same pixels twice and cannot memorise individual images; it is pushed instead toward features that survive these transformations.

These particular transformations are valid because a cell has no canonical orientation. A nucleus rotated or mirrored is still the same nucleus of the same type, so the label is preserved, and the brightness and contrast jitter imitates the real variation in staining intensity between slides.

The effect on the loss curves is exactly what the theory predicts. The validation loss now falls in step with the training loss instead of turning upward, it minimizes the overfitting. The accuracy gain is modest, with macro-F1 rising to 0.878 on Task A and 0.646 on Task B, because augmentation adds no new information and cannot raise the ceiling of what is learnable from 27×27 patches. This

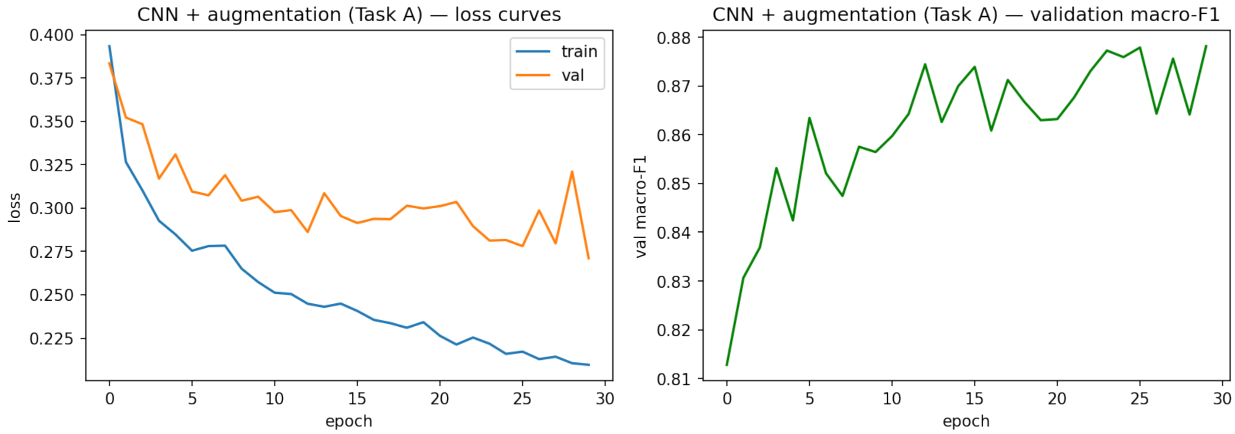

Figure 10: Task A CNN with augmentation, loss curves

is the trustworthy approach because a reduction in variance at a small cost in training accuracy. Especially, the augmented CNN is the best single model on *both* tasks.

E.2 Ensemble by soft voting

The second technique averages the predicted class probabilities of three models: the MLP, the tuned CNN and the augmented CNN. That these models carry different inductive biases, the MLP seeing global pixel patterns and the CNNs seeing local spatial structure, so they should make partly independent mistakes. Averaging predictors whose errors are decorrelated cancels some of that error and reduces variance without adding bias, which is the standard justification for ensembling.

It did not work. On Task A the ensemble scored 0.871, below the augmented CNN; on Task B it dropped to 0.604, below even the plain CNN, and the rare *other* class got worse rather than better. The reason is a condition that the standard justification quietly assumes: the members must be not only diverse but comparably accurate. The MLP is much weaker than the CNNs (0.826 against 0.867 on Task A, and 0.534 against 0.635 on Task B), so giving it an equal vote pulls the average toward its mistakes. The bias injected by the weak member outweighs the variance removed by averaging.

This negative result shows that ensembling is not met in this scenario.

F. Model Comparison

Model	Task A macro-F1	Task B macro-F1
Decision Tree	0.729	0.427
Logistic Regression	0.799	0.492
MLP	0.826	0.534
CNN	0.867	0.635
CNN + augmentation	**0.878**	**0.646**
Ensemble (soft vote)	0.871	0.604

The table shows a clean story up to the CNN. Each increase in capacity buys an increase in

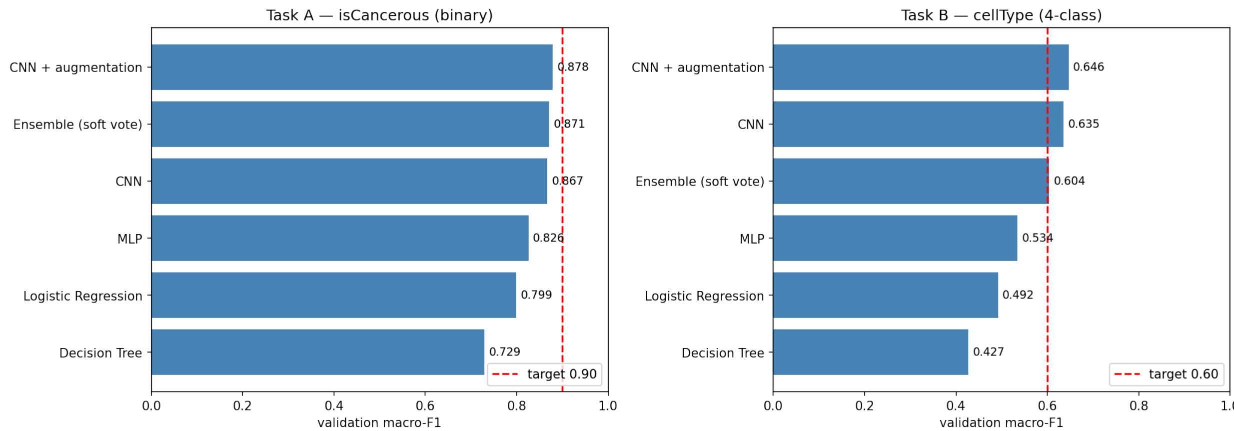

Figure 11: Validation macro-F1 for every model, both tasks

performance, from the linear models, through the non-linear but spatially blind MLP, to the spatially aware CNN. This is exactly the ordering the EDA predicted: because the classes are not linearly separable in pixel space, the models that can learn richer, local features win.

Past the CNN, though, capacity stops being the lever. Augmentation gives the best model on both tasks, but it does so by regularising the CNN rather than by adding capacity, and the ensemble, which is the most complex option of all. The lesson is that complexity helps only when the model is already expressive enough. Then the remaining gains come from better generalisation which does not from more parameters.

The augmented CNN is therefore the final model for both tasks. It is the strongest on validation, and it is a single interpretable model rather than a three-model committee that bought nothing.

G. Final Model Selection and Evaluation

Having chosen the augmented CNN on the basis of validation performance, I evaluated it once on the held-out test patients. This is the only time the test set is used, so the figures below are an unbiased estimate of how the model would perform on genuinely new patients.

Task A

Task A generalises almost perfectly. Validation and test macro-F1 are 0.878 and 0.876, a gap of just 0.002, and the model reaches 90% accuracy while correctly identifying 89% of cancerous cells. That tiny gap is the reward for patient-level splitting, so it confirms the validation score was honest and that the model learned transferable features rather than specific patients.

The model nonetheless lands a little below the 0.90 macro-F1 target. I read this as a real ceiling rather than a failure of tuning. At 27×27 pixels some cells are genuinely ambiguous, the EDA already showed that cancerous and non-cancerous cells overlap in pixel space, and an small error follows from that overlap no matter how the model is trained. The remaining error is also the safer kind to skew, since the model's recall on cancerous cells (0.888) is higher than its precision (0.767), meaning it errs toward flagging rather than missing.

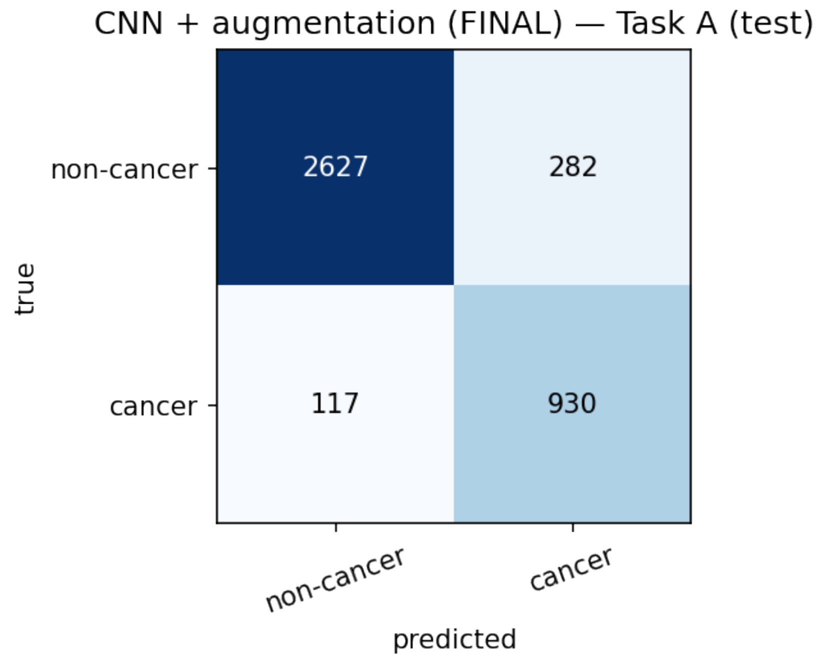

Figure 12: Task A final test confusion matrix

Task B

Task B tells the opposite story and the finding is surprising. Validation macro-F1 was acceptable with 0.646, but on held-out patients it collapses to 0.475, and a gap of 0.17 to the target.

The first is that the validation score was optimistic from the start. I chose the augmented CNN and its learning rate by reading the validation numbers, that score reflects a model partly fitted to the quirks of those particular validation patients. The test set was used for no decision at all, so the drop from 0.646 to 0.475.

The second, and the deepest, is the nature of the question. Task A relies on coarse, broadly universal cues, since cancerous epithelial cells are large and dark in a way that looks similar across patients, and that is why its validation and test scores are almost identical. Task B requires more how to separate fibroblast from *other* from inflammatory on the basis of subtle shape and texture that genuinely differs from one patient to the next with tissue, staining batch and scanner. The model learned what those types look like in the *training* patients, and that knowledge did not transfer.

The third effect is that the rare classes live in very few patients. The patient-level split leaves roughly thirty-six training patients, and the rarest classes, *other* with 1,386 cells and fibroblast with 1,888, are spread across only a handful of them. With so few patients carrying these cells, the model cannot learn much from them. It learns what fibroblasts looked like in those specific patients. When the test patients' rare cells differ even slightly, there is no general rule to fall back on. Only the common epithelial class, the same cells as the cancerous label, stays reliable at an F1 of 0.79.

The fourth effect multiplies the third. Macro-F1 averages the four classes equally, by design, so that a model cannot hide a neglected minority behind the common classes. That is the correct metric for an imbalanced problem, but it also means that when the two rare classes collapse on test, they drag the whole score down out of proportion to their size. On validation those rare classes happened to

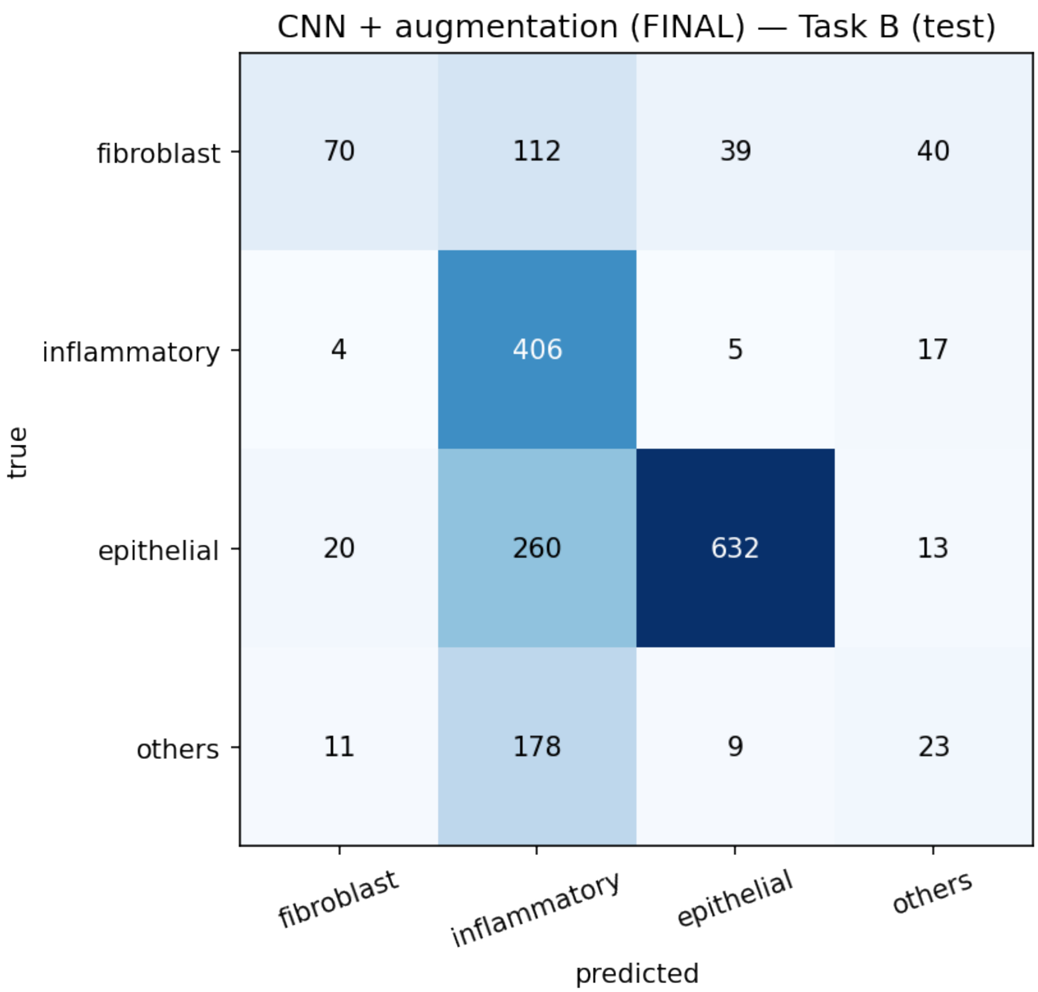

Figure 13: Task B final test confusion matrix

do passably, because the validation patients' rare cells resembled the training ones, but they did not on test, and macro-F1 turned that into a large fall.

H. Ethical Considerations and Professional Responsibilities

Although the data is de-identified it remains sensitive medical information. I have restricted its use to this coursework and have not redistributed it.

The concern is what these models would mean if anyone deployed them. Neither is a medical device, and the test results show why that distinction matters. On Task A the model still misses about one cancerous cell in nine, and a missed cancer is the most costly error a screening tool can make. Task B does not transfer to new patients at all. At best a tool like this could assist a pathologist by flagging cells for a second look.

The models also inherit the limits of their data. Ninety-eight patients from a single source represent a narrow slice of the variation found in practice, where staining protocols, scanners and patient populations all differ, and there is no basis for assuming the model behaves fairly on groups it never saw. Task B's collapse on unseen patients is direct, measured evidence of that fragility.

Professional responsibility, finally, means reporting these limits plainly rather than quoting the kinder validation figures. I report Task B's honest test score of 0.475, and I chose a single CNN over the ensemble partly because one model is easier to scrutinise than a committee, while acknowledging that any CNN remains difficult to interpret and would need careful analysis and human oversight before it came near a clinic.

I. Conclusion

This project compared models of increasing capacity on two histopathology tasks under a strict patient-level evaluation. The EDA predicted, and the results confirmed, that convolutional models would outperform linear and flat ones because the classes are not linearly separable in pixel space. Data augmentation produced the best single model on both tasks by curing the CNN's overfitting rather than by adding capacity, while the soft-voting ensemble failed because its weakest member dragged the average down, a reminder that ensembling requires members of comparable skill.

The decisive lesson came from the test set. Task A generalised almost perfectly, reaching a test macro-F1 of 0.876 just short of the 0.90 target, but Task B fell from 0.646 on validation to 0.475 on unseen patients. Validation alone would have concealed this, and only patient-level evaluation revealed that the cell-type model does not transfer between patients. The honest conclusion is that the binary cancer task is close to solved at this resolution, while reliable cell typing will need class-balanced training and collect more data from patients.

J. References

Practical Labs in this course.